CMR INSTITUTE OF TECHNOLOGY
(UGC AUTONOMOUS)
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DEPARTMENT OF CSE,CSE(AIML),CSE(DS)

Assignment I

CourseName	**:**	**Operating System**
CourseCode	**:**	**24CSPC43**
Class-Sem	**:**	**II year–II Sem(R24–CMRIT)**
Branch	**:**	**CSE,CSE(AIML),CSE(DS)**
Year	**:**	**2025-2026**
Course Faculty	**:**	**Mrs. A.Hemalatha Reddy**

S.No	SET-1Question	BTL	PO/PSO	CO
1	a) Describe the services of the Operating System. b)Explain various functions of Operating System.	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	1
2	a) Explain any 3 types of operating systems with their merits and demerits. b) Illustrate about different types of system calls in OS.	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	1
3	a) What is a process?Explain different states of a process. b) Consider the following set of process with the arrival time,with the length of the CPU burst given in milliseconds Process Arrival Time Bursttime P1 0 13 P2 5 22 P3 8 5 P4 2 11 Calculate the average waiting time using SJF,RR (TS=4)	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	2
4	a) Explain in detail about IPC mechanism. b) Explain about Fork system call in detail.	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	2
5	a) Explain deadlock recovery methods. b) What are the conditions to be met for the occurrence of a deadlock?	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	3

S.No	SET-2Question	BTL	PO/PSO	CO
1	a) List the essential properties of the following types of operating systems. i)batch ii)Distributed iii)timesharing iv)realtime	IV	PO1,2,3,4,5,6,8,9,11,PSO1,2	1
2	a) Classify the different operating system structures b)Define system call and list the various types of system calls.	I,IV	PO1,2,3,4,5,6,8,9,11,PSO1,2	1
3	a) Explain the schedulers and scheduling queues used in CPU scheduling. b) What is meant by context switching?	II	PO1,2,3,4,5,6,8,9,11,PSO1,	2

4	Explain reader writer problem and provide solution using semaphores.	II	PO1,2,3,4,5,6,8,9,11,PSO1,2	2
5	a) Define deadlock and give a real-time example of deadlock situation. b) Illustrate Banker's algorithm with proper example.	I,II	PO1,2,3,4,5,6,8,9,11,PSO1,2	3

S.No	SET-3Question	BTL	PO/PSO	CO
1	a) Summarize the list of evolution of Operating Systems. b) List the goals of Operating system.	I,II	PO1,,2,3,4,5,6,8,9,11,PSO1,2	1
2	a)Differentiate between Multiprogramming,Multiprocessing and Multitasking. b) Label the computer system architecture?Represent the different views of a system.	IV	PO1,,2,3,4,5,6,8,9,11,PSO1,2	1
3	a) Illustrate ProgramControlBlock (PCB) indetail. Consider the following set of process with the arrival time,with the length of the CPU burst given in milliseconds Process Arrival Time Bursttime Priority P1 0 25 3 P2 5 10 2 P3 10 18 1 P4 15 12 4 b) Calculate the average waiting time using FCFS,Priority(non preemptive).	II	PO1,,2,3,4,5,6,8,9,11,PSO1,2	2
4	Define critical section problem and explain its software and hardware solutions.	III	PO1,,2,3,4,5,6,8,9,11,PSO1,2	2
5	a) Explain how a deadlock is evaluated by Resource Allocation Graph(RAG). b) Explain the necessary conditions for deadlocks.	II	PO1,,2,3,4,5,6,8,9,11,PSO1,2	3

S.No	SET-4Question	BTL		CO
1	a) Define OS? What are the various objectives and functions of the Operating Systems? b) Explain the simple and layered approach of operating system design in detail.	I, II	PO1,,2,3,4,5,6,8,9,11,PSO1,2	1
2	a) Differentiate between Multi programming and Multiprocessing b) List some Linux utilities with syntaxes.	IV	PO1,,2,3,4,5,6,8,9,11,PSO1,2	1
3	a) Distinguish between a process and thread. b) Explain fork and vfork system calls indetail.	IV	PO1,,2,3,4,5,6,8,9,11,PSO1,2	2
4	Illustrate classic problem of synchronization.	III	PO1,,2,3,4,5,6,8,9,11,PSO1,2	2
5	a) Explain the methods for handling a deadlock. b) State and explain the steps involved in safety algorithm.	II	PO1,,2,3,4,5,6,8,9,11,PSO1,2	3